

Project Plan

B.Tech CSE 2025–29 · Software Engineering & Project Management · Semester III

Dashboard basis: The project author has developed three dashboards—Student, Faculty and Admin. This document uses the supplied walkthrough to describe them and distinguishes those dashboard details from requirements, integrations or metrics that are proposed or not independently verified.

IN THIS DOCUMENT

1. Planning decision • 2. Work breakdown structure • 3. Network diagram and forward pass • 4. Gantt-style schedule • 5. Resource allocation • 6. Milestones and exit criteria • 7. Schedule controls

1. Planning decision

The platform is stated to be ready in 10 weeks, while the full 30-course content catalogue is estimated to require 18 weeks at the stated editing capacity. The recommended plan is therefore a **phased release**: make the platform and the first 8 prepared courses available when both are ready (planning week 10), then add the remaining 22 courses as they pass quality review (planning week 18). A 30-course launch in week 10 is not supported by the supplied production calculation.

The 4.8-week figure for 8 courses is an effort/capacity estimate, not a firm faculty delivery promise. The problem statement says faculty can record only 8 courses in the next three months; the actual recording calendar must be agreed with faculty. The plan uses week numbers from project start rather than calendar dates because no start date was provided.

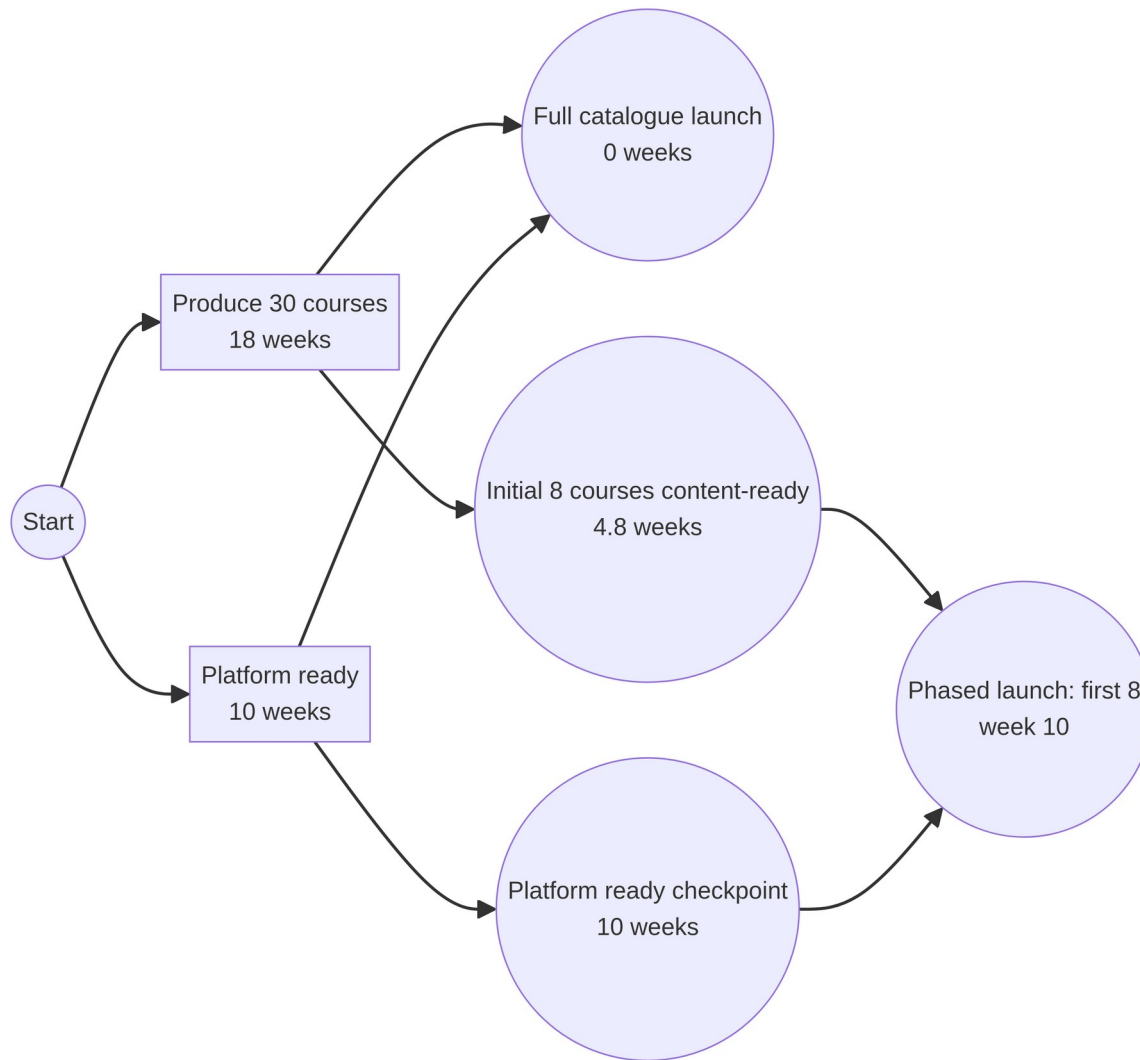
2. Work breakdown structure

WBS	Work package	Main output	Owner / contributors
1.0	Initiation and requirements	Approved scope, stakeholder needs, MoSCoW backlog	Director, analyst, faculty, learners, IT
1.1	Stakeholder workshop and document analysis	Validated requirements and assumptions	Analyst + stakeholder representatives
1.2	Architecture and UX design	API boundaries, role flows, design and UML	Architect, UX, IT
2.0	Platform implementation	Student, faculty and admin dashboards; backend services	Development team
2.1	Identity and role access	Registration, sign-in and access controls	Development + security review

WBS	Work package	Main output	Owner / contributors
2.2	Course, enrolment and payment workflow	Catalogue, enrolment records and verified payment adapter	Development + finance/payment owner
2.3	Learning, quiz and certificate workflow	Lesson progress, assessment, certificate eligibility	Development + academic owner
2.4	Dashboard and reporting	Student/faculty/admin views and metrics	Development + stakeholder review
2.5	Integration, testing and release readiness	Test evidence, defect triage, deployment checklist	QA + IT + business owner
3.0	Course content production	30 edited, reviewed course packages	Faculty and 2 editors
3.1	Confirm course list and faculty slots	Approved production sequence	Faculty coordinator
3.2	Record and edit initial 8 courses	First release content	Faculty + editors
3.3	Record and edit remaining 22 courses	Full catalogue content	Faculty + editors
3.4	Academic and media quality checks	Approved videos, captions and lesson metadata	Faculty + QA/editor
4.0	Release and operations	Pilot release, monitoring, support and lessons learned	IT, Director, faculty

3. Network diagram and forward pass

For the case's launch comparison, the schedule model uses two parallel workstreams from project start: platform readiness (10 weeks, as stated) and full-catalogue content production (calculated as 18 weeks). The full-catalogue launch checkpoint depends on both. The initial 8-course milestone is shown separately for the staged-release decision.



Project network diagram

Forward pass / critical path

Activity	Duration	Predecessor	ES	EF	LS	LF	Total float	Status
A. Platform ready	10 weeks	Start	0	10	8	18	8 weeks	Non-critical to full-catalogue launch

Activity	Duration	Predecessor	ES	EF	LS	LF	Total float	Status
B. Produce and approve 30 courses	18 weeks	Start	0	18	0	18	0 weeks	Critical path
C. Full catalogue launch checkpoint	0 weeks	A, B	18	18	18	18	0	Milestone

Forward-pass rule: $EF = ES + \text{duration}$; where predecessors join, ES is the maximum predecessor EF. Thus full-catalogue launch is at $\max(10, 18) = \text{week 18}$. The **critical path is Start → B (30-course content production) → C**. The platform workstream has **8 weeks total float** ($18 - 10$) before it delays the full-catalogue milestone.

For the staged release, the first 8 courses are estimated ready after $8 \times 36 \div 60 = \text{4.8 weeks}$. The platform is ready at week 10, so the initial staged-release checkpoint is at $\max(4.8, 10) = \text{week 10}$. The additional 22 courses need $22 \times 36 \div 60 = \text{13.2 further weeks}$ at the same uninterrupted capacity, bringing full content readiness to week 18. The remaining-content work is on the content critical path. These are estimates, not commitments.

4. Gantt-style schedule

Each ♦ milestone is a zero-duration checkpoint, not a duration-bearing task.

Workstream / activity	W0-2	W2-4	W4-6	W6-8	W8-10	W10-12	W12-14	W14-16	W16-18	Milestone
Requirements and design (within 10-week platform window)	■									
Platform build, integration and QA (overall readiness by W10)	■	■	■	■	■					♦ Platform ready W10
Produce initial 8 courses (4.8-week effort estimate)	■	■	■							♦ 8 courses content-ready ~W4.8
Phased initial release (gated by platform and first 8 courses)										♦ W10

Workstream / activity	W0-2	W2-4	W4-6	W6-8	W8-10	W10-12	W12-14	W14-16	W16-18	Milestone
Produce remaining 22 courses (13.2-week effort estimate)			■	■	■	■	■	■	■	◆ Full content ready W18
Full catalogue release										◆ W18

The bars are indicative; overlapping content work assumes faculty access and editor capacity can be sequenced without blocking. No calendar start date or contingency is supplied, so none is implied.

5. Resource allocation

Resource	Planning allocation	Main responsibility / constraint
2 editors	30 hours/week each; 60 hours/week combined	Editing and production effort; planned as a capacity ceiling
Faculty	Course-dependent; case says only 8 courses can be recorded in next 3 months	Recording, academic review and question-bank approval; availability is a major risk
Development team	Sized by university; platform total window = 10 weeks	Build, integration, security and release readiness
QA / IT operations	Involve throughout; no headcount supplied	Test execution, access review, monitoring, backup and release
Director / admin	Review at gates	Scope, staged-release decision, revenue targets and acceptance

6. Milestones and exit criteria

Milestone	Target week	Exit criteria
M1 — Requirements baseline	By W2 (planning assumption within platform window)	Stakeholder needs and open decisions recorded; FR/NFR acceptance targets agreed
M2 — Initial content-ready	~W4.8 estimate	Eight courses have approved video, lesson metadata and quiz content
M3 — Platform ready	W10 (case-stated target)	Core test suite passes; payment status verified; role access and backup checks complete
M4 — Phased first release	W10	Platform ready and initial 8 content packages approved; support owner identified

Milestone	Target week	Exit criteria
M5 — Full catalogue content-ready	W18 estimate	All 30 course packages pass academic and media review
M6 — Full catalogue release	W18	Full catalogue available; operational metrics and escalation process active

7. Schedule controls

Review production status weekly; record planned versus actual faculty sessions and editor hours. Reforecast when a course slips by more than one week, when combined editing capacity falls below 60 hours/week, or when fewer than eight courses are on track for the planned initial release. The Director should approve changes to release scope; adding work without moving a date is not assumed.